Math Bootcamp

Day 3 — Remaining real analysis and linear algebra

Math Camp 2026

How to use these slides

Main deck: definitions, recipes, examples, and practice.

Appendix and proofs (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

Not required for Math Camp. Focus on the main slides and exercises unless you want the extra detail.

3.1 Fixed points — equilibrium as a solution

Fixed point. A point x^* is a **fixed point** of a map T if $T(x^*) = x^*$.

Economics reading. Many equilibria are fixed-point problems:

- **Market clearing:** a price p^* with excess demand $E(p^*) = 0$.
- **Game theory:** a strategy profile that is a best response to itself (Nash).

Two useful existence tools (today).

- **Contraction Mapping Theorem:** existence and uniqueness of a fixed point; iteration converges.
- **Brouwer:** continuous map on a closed interval / convex set.

3.1 Contraction Mapping Theorem

Contraction. A map T **shrinks distances** if there is $k \in (0,1)$ such that $|T(x) - T(y)| \le k|x - y|$ for all x, y in the domain.

Contraction Mapping Theorem. If $T\colon [a,b] \to [a,b]$ is a contraction, then

- **Existence:** there is a fixed point x^* with $T(x^*) = x^*$.
- **Uniqueness:** that fixed point is **unique**.
- **Iteration:** for any starting value $x_0 \in [a,b]$, the sequence $x_n = T(x_{n-1})$ converges to x^*.

3.1 Why iteration converges

Step 1 — Start anywhere. Pick any x_0. Define $x_n = T(x_{n-1})$ for $n = 1, 2, \ldots$

Step 2 — Jumps shrink exponentially. Each new step is at most k times the previous jump:

$$|x_{n+1} - x_n| = |T(x_n) - T(x_{n-1})| \leq k|x_n - x_{n-1}| \leq k^n|x_1 - x_0|.$$

Since $k < 1$, the moves get **exponentially smaller** — the path barely changes after a while.

Step 3 — The sequence converges. Step 2 implies $x_n \to x^*$ for some limit x^* (check it is a **Cauchy sequence** for a rigorous proof). So iteration from **any** starting point drives x_n toward the same x^*.

Step 4 — The limit is a fixed point. T is continuous, so taking limits in $x_n = T(x_{n-1})$ gives $x^* = \lim x_n = \lim T(x_{n-1}) = T(\lim x_{n-1}) = T(x^*)$.

3.1 Walrasian price adjustment

Market clearing. A price p^* **clears the market** when excess demand is zero: $E(p^*) = 0$. Here $E(p)$ can take **any form** (from supply and demand behind the scenes).

Price adjustment (nonlinear in E). Update the price each period:

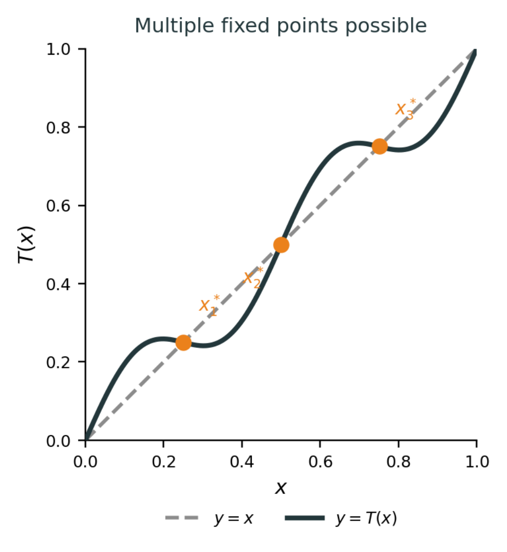
$$p_{t+1} = T(p_t) = p_t + \lambda\big(1 - e^{-\alpha E(p_t)}\big), \qquad \lambda = \tfrac{1}{5},\ \alpha = 0.1.$$

When $|E|$ is small, $1 - e^{-\alpha E} \approx \alpha E$ — move a fraction of excess demand toward balance. Clearing satisfies $E(p^*) = 0 \iff T(p^*) = p^*$. If T is a **contraction** on a price range, then $p_t \to p^*$ from any starting p_0.

3.1 Brouwer — one dimension

Brouwer (one dimension). If $T\colon [a,b] \to [a,b]$ is **continuous**, then T has **at least one** fixed point x^* with $T(x^*) = x^*$.

Picture. Plot $y = T(x)$ against $y = x$. Any crossing is a fixed point. Continuity forces the graph to meet the diagonal at least once on $[a,b]$.


Multiple fixed points possible
1.0
0.8
0.6
0.4
0.2
0.0
T(x)
0.0
0.2
0.4
0.6
0.8
1.0
x
x_1^*
x_2^*
x_3^*
$y = x$
$y = T(x)$

3.1 Brouwer — general statement

Brouwer (several variables). Let $K \subset \mathbb{R}^n$ be **nonempty**, **compact**, and **convex**. If $T : K \to K$ is **continuous**, then T has at least one fixed point $x^* \in K$ with $T(x^*) = x^*$.

Camp vocabulary.

- **Compact** in $\mathbb{R}^n$: closed and bounded (Heine–Borel).
- **Convex**: the segment between any two points in K stays in K.

One dimension. $K = [a, b]$ is compact and convex, so the 1D case is a special case.

3.1 Brouwer — Cournot duopoly

Two firms. Firm i chooses $q_i \in [0, \bar{q}]$. **Nonlinear** inverse demand $p = a - (q_1 + q_2)^2$ (e.g. $a = 8$, $\bar{q} = 2$).

Profit and FOC. $\pi_i = (a - (q_1 + q_2)^2)q_i$. Firm 1's best response solves

$$a - 3q_1^2 - 4q_1q_2 - q_2^2 = 0 \qquad \text{(similarly for firm 2 with } q_1, q_2 \text{ swapped).}$$

The map $q_j \mapsto BR_i(q_j)$ is **nonlinear** — no formula like "$(a - c - q_j)/2$".

Fixed point = Nash. $T(q_1, q_2) = \big(BR_1(q_2), BR_2(q_1)\big)$ is continuous on $K = [0, \bar{q}]^2$. **Brouwer** guarantees an equilibrium, but you generally solve $T(q_1^*, q_2^*) = (q_1^*, q_2^*)$ by **algebra or numerics** (here one symmetric solution is $q_1^* = q_2^* = \sqrt{a/8}$).

3.2 Separating Hyperplane Theorem

Setup. Two nonempty **convex** sets $A, B \subset \mathbb{R}^n$ with $A \cap B = \varnothing$.

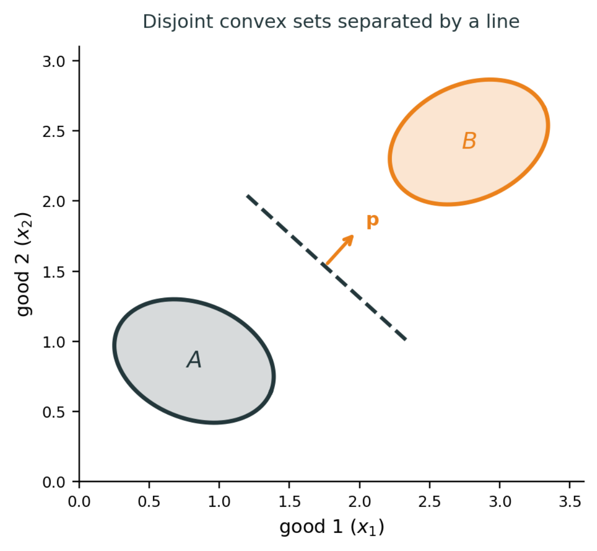
Theorem. If **at least one** of A, B is **compact**, then there exist $\mathbf{p} \neq \mathbf{0}$ and $\beta \in \mathbb{R}$ such that $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ for all $a \in A$, $b \in B$. Geometrically: the hyperplane $\mathbf{p} \cdot x = \beta$ separates A from B. (Both need not be compact — if only B is compact, swap labels.)

In two goods ($x = (x_1, x_2)$). The line $p_1 x_1 + p_2 x_2 = \beta$ puts all of A on one side ($\mathbf{p} \cdot a \leq \beta$) and all of B on the other ($\mathbf{p} \cdot b \geq \beta$); $\mathbf{p}$ is the normal, β fixes the line's position.

Remark (single point). $B = \{b_0\}$ is compact. If $b_0 \notin A$ and A is closed and convex, a separating hyperplane exists even when A is not compact.

Remark (strict separation). If the sets are a **positive distance** apart ($d = \inf \|a - b\| > 0$; e.g. both compact and disjoint), the same $\mathbf{p}$ gives **strict** separation $\mathbf{p} \cdot a < \beta < \mathbf{p} \cdot b$ — appendix (**for interest only**).

Proof idea when A is compact: appendix (**for interest only**).


Disjoint convex sets separated by a line
3.0
2.5
2.0
1.5
1.0
0.5
0.0
good 2 (x_2)
B
p
A
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
good 1 (x_1)

3.2 Example — producer theory (numerical illustration)

Netput plan $y = (y_1, y_2)$: input $y_1 \leq 0$, output $y_2 \geq 0$ (second quadrant).

Technology. With $L = -y_1$ units of labor, max output is $f(L) = 2\sqrt{L}$, so

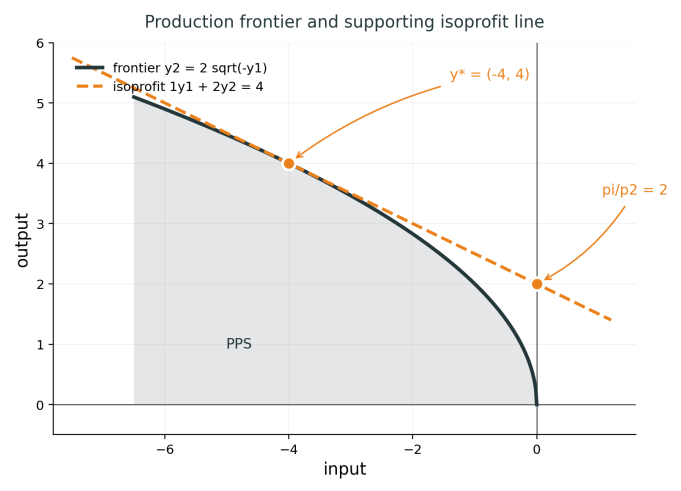
$$Y = \{(y_1, y_2) : \; y_1 \leq 0, \; 0 \leq y_2 \leq 2\sqrt{-y_1}\} \quad \text{(free disposal on output).}$$

Prices. Wage $p_1 = 1$, output price $p_2 = 2$. Profit $\pi = p_1 y_1 + p_2 y_2$ (since $y_1 < 0$, the term $p_1 y_1$ is labor cost).

Efficient plan. Use $L = 4$: $y^* = (-4, 4)$. Then $\pi = 1 \cdot (-4) + 2 \cdot 4 = 4$.

Isoprofit line. $y_1 + 2y_2 = 4$, or $y_2 = 2 - \frac{1}{2} y_1$. At $y_1 = 0$: y_2**-axis intercept** $y_2 = \pi / p_2 = 2$. Slope $-p_1/p_2 = -\frac{1}{2}$ matches the frontier at y^* (supporting prices).

3.2 Example — producer theory (graph)



3.2 Example — producer theory

Setup. A firm's **production possibility set** $Y \subset \mathbb{R}^n$ is **convex** (and closed in full treatments).

Statement. If Y is convex, the theorem guarantees that every **technically efficient** production plan $y \in Y$ is also a **profit-maximizing** choice for some nonzero price vector $\mathbf{p}$:

$$\mathbf{p} \cdot y \geq \mathbf{p} \cdot y' \quad \text{for all } y' \in Y.$$

Separation picture. Let B be outputs **strictly better than** y (more of some good, no less of any). If y is efficient, $Y \cap B = \varnothing$. A separating hyperplane gives supporting prices $\mathbf{p}$ at y.

Proof: appendix (**for interest only**).

Linear Algebra Basics

Vector. An ordered list of n real numbers. We write a **column vector** as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Its **transpose** $\mathbf{x}^\mathsf{T} = (x_1, \dots, x_n)$ is a **row vector**.

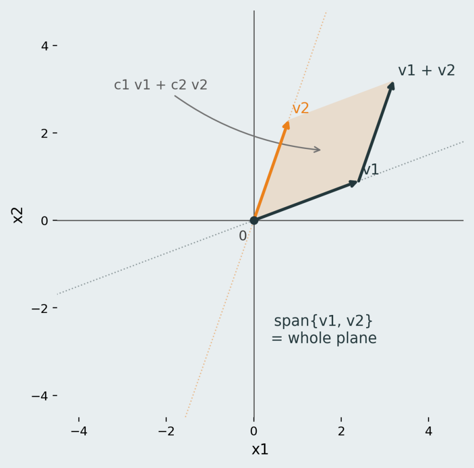
Linear combination. Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and scalars $c_1, \dots, c_k$,

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$$

is a **linear combination** of the $\mathbf{v}_j$.

Span. All linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_k$ form the **span** of those vectors.

3.3 Span in the plane (two vectors)



3.3 Matrix–vector product — columns of A

Setup. $A \in \mathbb{R}^{m\times n}$ has columns $\mathbf{a}_1, \ldots, \mathbf{a}_n \in \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$ is a column vector.

Column view. $A\mathbf{x}$ is a linear combination of the **columns** of A:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n \in \mathbb{R}^m.$$

The weights are the entries of $\mathbf{x}$.

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$, $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$:

$$A\mathbf{x} = x_1 \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}.$$

3.3 Row vector times matrix — rows of A

Setup. $A \in \mathbb{R}^{m\times n}$, $\mathbf{x} \in \mathbb{R}^n$ (column), so $A\mathbf{x} \in \mathbb{R}^m$.

Row view (transpose both sides). $(A\mathbf{x})^\mathsf{T} = \mathbf{x}^\mathsf{T} A^\mathsf{T}$:

$$\mathbf{x}^\mathsf{T} A^\mathsf{T} = x_1\,\mathbf{a}_1^\mathsf{T} + x_2\,\mathbf{a}_2^\mathsf{T} + \cdots + x_n\,\mathbf{a}_n^\mathsf{T} \in \mathbb{R}^{1\times m}.$$

Dimensions: $(1\times n)(n\times m) = (1\times m)$. **Note:** $\mathbf{x}^\mathsf{T} A$ does **not** match when $A \in \mathbb{R}^{m\times n}$ with $m \neq n$.

Example. From the previous slide,

$$\mathbf{x}^\mathsf{T} = (x_1, x_2), \qquad A^\mathsf{T} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}.$$

Then

$$\mathbf{x}^\mathsf{T} A^\mathsf{T} = x_1(1,3,5) + x_2(2,4,6).$$

3.3 Two views of the same product

Same numbers, two interpretations. If $A \in \mathbb{R}^{m\times n}$ and $\mathbf{x} \in \mathbb{R}^n$:

- **Column view:** $A\mathbf{x} = \sum_{j=1}^{n} x_j\, \mathbf{a}_j \in \mathbb{R}^m$.
- **Entry view:** $(A\mathbf{x})_i = \mathbf{r}_i^\top A\mathbf{x} = \sum_{j=1}^{n} a_{ij}x_j$ (dot product of row i of A with $\mathbf{x}$), where $\mathbf{r}_i \in \mathbb{R}^m$ has i-th entry 1 and all other entries 0 ($\mathbf{r}_i = \mathbf{e}_i$).

Transpose link. $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$: a **row vector** on the left combines **rows of** $A^\top$ (i.e. transposes of columns of A).

3.3 Matrix multiplication AB — example

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$.

Column view (both columns of AB).

$$(AB)\mathbf{e}_1 = A\begin{pmatrix} 5 \\ 7 \end{pmatrix} = 5\begin{pmatrix} 1 \\ 3 \end{pmatrix} + 7\begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 19 \\ 43 \end{pmatrix},$$

$$(AB)\mathbf{e}_2 = A\begin{pmatrix} 6 \\ 8 \end{pmatrix} = 6\begin{pmatrix} 1 \\ 3 \end{pmatrix} + 8\begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 22 \\ 50 \end{pmatrix}.$$

So $AB = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$.

Row view (practice). Find rows 1 and 2 of AB using $(1,2)B$ and $(3,4)B$.

Note (skip in class). $(1,2)B = (19,22)$ and $(3,4)B = (43,50)$ — same matrix as above.

3.3 Matrix multiplication AB

Setup. $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, so $AB \in \mathbb{R}^{m\times p}$.

Column view. Column j of AB is A times column j of B:

$$(AB)\,\mathbf{e}_j = A(B\mathbf{e}_j) = A\mathbf{b}_j = b_{1j}\mathbf{a}_1 + b_{2j}\mathbf{a}_2 + \cdots + b_{nj}\mathbf{a}_n.$$

Each column of AB is a linear combination of **columns of** A (weights from column j of B).

Row view. Row i of AB is row i of A times B:

$$\mathbf{r}_i^{\mathsf{T}}(AB) = (\mathbf{r}_i^{\mathsf{T}}A)B = a_{i1}\mathbf{s}_1^{\mathsf{T}} + \cdots + a_{in}\mathbf{s}_n^{\mathsf{T}},$$

where $\mathbf{r}_i \in \mathbb{R}^m$ has i-th entry 1 and all other entries 0 ($\mathbf{r}_i = \mathbf{e}_i$), so $\mathbf{r}_i^{\mathsf{T}}$ selects row i; and $\mathbf{s}_k^{\mathsf{T}}$ is row k of B: a linear combination of **rows of** B.

3.3 Basis

Idea. A **basis** of $\mathbb{R}^n$ is a small set of vectors from which every vector in $\mathbb{R}^n$ can be built as a **linear combination** in exactly one way.

Standard basis of $\mathbb{R}^n$. For $j = 1, \ldots, n$, define $\mathbf{e}_j$ to have a 1 in entry j and 0 elsewhere:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Coordinates. Every $\mathbf{x} \in \mathbb{R}^n$ can be written uniquely as

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n,$$

where $x_1, \ldots, x_n$ are the **coordinates** (entries) of $\mathbf{x}$.

3.3 Basis — example ($\mathbb{R}^2$)

Key point. A basis is **not unique**, but for a **fixed** basis every vector has a **unique** representation as a linear combination.

Two bases of $\mathbb{R}^2$. Standard: $\mathbf{e}_1 = (1, 0)$, $\mathbf{e}_2 = (0, 1)$. Alternative: $\mathbf{v}_1 = (1, 0)$, $\mathbf{v}_2 = (1, 1)$.

Same vector, two representations. For $\mathbf{b} = (2, 3)$:

$$\mathbf{b} = 2\mathbf{e}_1 + 3\mathbf{e}_2 \qquad \text{and} \qquad \mathbf{b} = (-1)\mathbf{v}_1 + 3\mathbf{v}_2.$$

Different bases give different weights $x_1, \dots, x_n$, but within one basis the weights are **unique**.

Requirement for a basis $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$. Every $\mathbf{b} \in \mathbb{R}^n$ can be written as $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**.

3.3 Basis — existence and uniqueness

Requirement for a basis $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$. Every $\mathbf{b} \in \mathbb{R}^n$ can be written as $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**.

Two parts.

- **Existence:** every $\mathbf{b} \in \mathbb{R}^n$ can be written as some linear combination of $\mathbf{a}_1, \ldots, \mathbf{a}_n$.
- **Uniqueness:** there is **only one** choice of weights $x_1, \ldots, x_n$.

For $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{m\times n}$. Then $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, and $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ with $\mathbf{a}_j \in \mathbb{R}^m$ gives

$$A\mathbf{x} = \mathbf{b} \quad \Longleftrightarrow \quad \mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

Setup. $A \in \mathbb{R}^{m\times n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ with $\mathbf{a}_j \in \mathbb{R}^m$.

Column space.

$$\mathrm{Col}(A) = \mathrm{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} \subseteq \mathbb{R}^m.$$

Existence of a solution. $A\mathbf{x} = \mathbf{b}$ has **at least one** solution $\iff$ $\mathbf{b} \in \mathrm{Col}(A)$.

Meaning. $\mathbf{b} \in \mathrm{Col}(A)$ means $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ for some $\mathbf{x}$ — the **existence** part of the basis requirement, but only for this one $\mathbf{b} \in \mathbb{R}^m$ (and only when $\mathrm{Col}(A) = \mathbb{R}^m$ does it hold for **all** $\mathbf{b}$).

3.3 Uniqueness — linearly independent columns

Setup. $A \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$.

Linearly independent. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are **linearly independent** if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \quad \Longrightarrow \quad c_1 = c_2 = \cdots = c_k = 0.$$

Linearly dependent: not independent — some nontrivial combination equals $\mathbf{0}$.

Uniqueness of a solution. If $A\mathbf{x} = \mathbf{b}$ has a solution, it is **unique** iff the columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ are **linearly independent**.

3.3 Linear independence $\Leftrightarrow$ unique solution

Claim. Fix $\mathbf{b} \in \mathbb{R}^m$. If $A\mathbf{x} = \mathbf{b}$ has **at least one** solution, then it has a **unique** solution iff $\mathbf{a}_1, \ldots, \mathbf{a}_n$ are linearly independent.

($\Rightarrow$) Independent $\Rightarrow$ unique. If $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$, then $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$, i.e.

$$(x_1 - y_1)\mathbf{a}_1 + \cdots + (x_n - y_n)\mathbf{a}_n = \mathbf{0}.$$

Independence gives $x_j - y_j = 0$ for all j, so $\mathbf{x} = \mathbf{y}$.

($\Leftarrow$) Dependent $\Rightarrow$ not unique. If the columns are dependent, some $\mathbf{d} \neq \mathbf{0}$ satisfies $A\mathbf{d} = \mathbf{0}$. If $\mathbf{x}$ solves $A\mathbf{x} = \mathbf{b}$, then so does $\mathbf{x} + \mathbf{d}$:

$$A(\mathbf{x} + \mathbf{d}) = A\mathbf{x} + A\mathbf{d} = \mathbf{b} + \mathbf{0} = \mathbf{b},$$

and $\mathbf{x} + \mathbf{d} \neq \mathbf{x}$. So the solution is **not** unique.

3.3 Linear independence — examples ($\mathbb{R}^2$)

Independent (bases).

- Standard: $\mathbf{e}_1 = (1, 0)$, $\mathbf{e}_2 = (0, 1)$.
- Alternative: $\mathbf{v}_1 = (1, 0)$, $\mathbf{v}_2 = (1, 1)$.

For each pair, $c_1\mathbf{w}_1 + c_2\mathbf{w}_2 = \mathbf{0}$ implies $c_1 = c_2 = 0$.

Dependent (too many vectors). $\mathbf{e}_1 = (1, 0)$, $\mathbf{e}_2 = (0, 1)$, and $\mathbf{u} = (1, 1)$:

$$\mathbf{u} = \mathbf{e}_1 + \mathbf{e}_2 \quad \iff \quad \mathbf{e}_1 + \mathbf{e}_2 - \mathbf{u} = \mathbf{0}.$$

Coefficients $(1, 1, -1)$ are not all zero, so these three vectors are **linearly dependent**. A basis of $\mathbb{R}^2$ has exactly 2 vectors, not 3.

3.3 Linear dependence — columns example

Example (columns of A). $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, columns $\mathbf{a}_1 = (1, 2)^\top$, $\mathbf{a}_2 = (2, 4)^\top$.

Same direction. $\mathbf{a}_2 = 2\mathbf{a}_1$, so the two columns lie on one line through the origin.

Nontrivial combination to 0.

$$2\mathbf{a}_1 - \mathbf{a}_2 = \mathbf{0} \iff 2\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Coefficients $(2, -1)$ are not both zero, so $\mathbf{a}_1, \mathbf{a}_2$ are **linearly dependent** (only one independent direction in $\mathrm{Col}(A)$).

3.3 Solving $A\mathbf{x} = \mathbf{b}$ — example

Example. $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, so $\mathrm{Col}(A) = \mathrm{span}\{(1,2)\}$ (a line in $\mathbb{R}^2$).

Solvable. $\mathbf{b} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3(1,2)$, so $A\mathbf{x} = \mathbf{b}$ has solutions (e.g. $\mathbf{x} = (1,1)^\mathsf{T}$; not unique).

Not solvable (existence fails). $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \notin \mathrm{Col}(A)$, so $A\mathbf{x} = \mathbf{b}$ has **no** solution.

Dependent columns (uniqueness fails). $\mathbf{a}_2 = 2\mathbf{a}_1$, so the columns are **linearly dependent**. When a solution exists, it is not unique (e.g. both $\mathbf{x} = (3,0)^\mathsf{T}$ and $\mathbf{x} = (1,1)^\mathsf{T}$ give $\mathbf{b} = (3,6)^\mathsf{T}$).

3.3 Full basis requirement — only when $m = n$

Goal. When does **every** $\mathbf{b} \in \mathbb{R}^m$ satisfy $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**?

Claim. For $A \in \mathbb{R}^{m \times n}$, this happens **only if** $m = n$.

Proof sketch.

- **Uniqueness** $\Rightarrow$ columns independent. In $\mathbb{R}^m$ at most m vectors can be independent, so $n \leq m$.
- **Existence for all b** $\Rightarrow \mathrm{Col}(A) = \mathbb{R}^m$. With only n columns, $\dim(\mathrm{Col}(A)) \leq n$, so $n \geq m$.
- Therefore $m = n$.

If $n < m$, columns cannot span $\mathbb{R}^m$; if $n > m$, n columns in $\mathbb{R}^m$ cannot be independent.

Now $A \in \mathbb{R}^{n\times n}$, $\mathbf{b} \in \mathbb{R}^n$. Requirement for a basis $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$: every $\mathbf{b} \in \mathbb{R}^n$ is $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**.

Equivalent to $A\mathbf{x} = \mathbf{b}$.

$$A\mathbf{x} = \mathbf{b} \quad \Longleftrightarrow \quad \mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

Columns form a basis of $\mathbb{R}^n$ $\Longleftrightarrow$ existence and uniqueness for **every b** $\Longleftrightarrow$ $A\mathbf{x} = \mathbf{b}$ has a **unique** solution for every $\mathbf{b} \in \mathbb{R}^n$.

3.3 Invertible matrices — equivalent conditions

Let $A \in \mathbb{R}^{n\times n}$. The following are **equivalent**:

1. A is **invertible**: $\exists\, A^{-1}$ with $AA^{-1} = A^{-1}A = I_n$.
2. For every $\mathbf{b} \in \mathbb{R}^n$, $A\mathbf{x} = \mathbf{b}$ has a **unique** solution $\mathbf{x} = A^{-1}\mathbf{b}$.
3. Columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ form a **basis** of $\mathbb{R}^n$.
4. Columns are **linearly independent** ($\mathrm{Col}(A) = \mathbb{R}^n$).
5. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.
6. Rows of A are linearly independent (equivalently, form a basis of $\mathbb{R}^n$).
7. $\det(A) \neq 0$.

Next: what $\det(A)$ means geometrically, and why (6)–(7) fit this list.

3.3 Determinant — area scaling (2×2)

Setup. $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, columns $\mathbf{a}_1 = (a, c)^\top$, $\mathbf{a}_2 = (b, d)^\top$.

Where the standard basis goes.

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \mathbf{a}_1, \qquad A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} = \mathbf{a}_2.$$

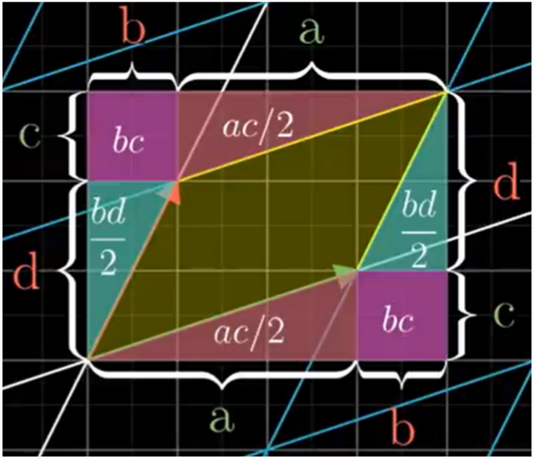
So $(1, 0)^\top$ is sent to $(a, c)^\top$, and $(0, 1)^\top$ to $(b, d)^\top$.

Area conversion factor. The unit square (spanned by $(1, 0)^\top$ and $(0, 1)^\top$, area $= 1$) is mapped by A to the parallelogram spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$. The **determinant** is exactly the (signed) area of that parallelogram:

$$\det(A) = ad - bc.$$

So $\det(A)$ is the **area conversion factor**: every region's area is multiplied by $|\det(A)|$ under $\mathbf{x} \mapsto A\mathbf{x}$.

3.3 Determinant — why $ad - bc$ is the area



Box area $(a+b)(c+d)$ minus purple (bc), red $(ac/2)$, teal $(bd/2)$ pieces leaves yellow parallelogram area

$$ad - bc = \det(A).$$

3.3 Determinant $= 0$ — example

Example. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, so $a = b = c = d = 1$ and

$$\det(A) = ad - bc = 1 \cdot 1 - 1 \cdot 1 = 0.$$

Standard basis collapses to a line.

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Both $(1, 0)^{\mathsf{T}}$ and $(0, 1)^{\mathsf{T}}$ are sent to the **same** point on one line through the origin. The unit square (area 1) is squashed to a **1-dimensional** set — area 0.

So (square case):

$$\det(A) = 0 \quad \iff \quad \text{columns linearly dependent} \quad \iff \quad A \text{ not invertible.}$$

Here $\mathbf{a}_2 = \mathbf{a}_1 = (1, 1)^{\mathsf{T}}$, so the columns are dependent and A has no inverse.

3.3 Two views of a matrix A

View 1 — $\mathbf{x}$ as weights. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

The entries of $\mathbf{x}$ are the **weights** on the columns of A.

View 2 — A as a linear transformation. Define $T : \mathbb{R}^n \to \mathbb{R}^m$ by $T(\mathbf{x}) = A\mathbf{x}$. The matrix A sends each input vector $\mathbf{x}$ to an output vector in $\mathbb{R}^m$; for square A, $|\det(A)|$ is the area-scaling factor (previous slides).

3.3 Linear transformation — definition

Linear transformation. A map $T : \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$:

$$T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}), \qquad T(c\mathbf{x}) = cT(\mathbf{x}).$$

Equivalently, for all $\mathbf{x}, \mathbf{y}$ and c:

$$T(c\mathbf{x}+\mathbf{y}) = cT(\mathbf{x}) + T(\mathbf{y}).$$

Examples. Rotation, reflection, projection, and shear in the plane are linear. A map with a constant term, e.g. $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$, is **not** linear.

Matrix form. Every linear T is $T(\mathbf{x}) = A\mathbf{x}$ for some matrix A (general recipe on the next slides). For $A \in \mathbb{R}^{2\times 2}$, $|\det(A)|$ is the area-scaling factor from the determinant slides.

3.3 Linear maps — decompose, then combine

Benefit. A linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ may be hard to visualize on an **arbitrary** input $\mathbf{v}$.

But you often know T on a few vectors. Suppose you already know $T(\mathbf{x})$ and $T(\mathbf{y})$, and $\mathbf{v}$ can be decomposed as

$$\mathbf{v} = a\mathbf{x} + b\mathbf{y}$$

for some scalars $a, b \in \mathbb{R}$. Then by linearity,

$$T(\mathbf{v}) = T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y}).$$

Decompose v into known pieces, **transform each piece**, then **combine**.

Standard basis recipe ($\mathbb{R}^2$). Take $\mathbf{x} = \mathbf{e}_1$, $\mathbf{y} = \mathbf{e}_2$, so $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$. If you know $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$, then $T(\mathbf{v}) = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2)$. Stack $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ as columns to get the matrix of T (next slide).

3.3 Decompose, transform, combine — notation

Recipe. For $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$ and linear T:

1. Decompose the input:

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2.$$

2. Transform each piece (use known values of T on $\mathbf{e}_1, \mathbf{e}_2$):

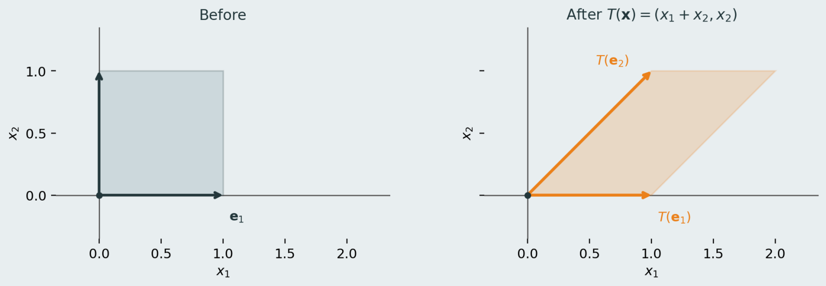
$$v_1 T(\mathbf{e}_1), \qquad v_2 T(\mathbf{e}_2).$$

3. Combine the outputs:

$$T(\mathbf{v}) = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) = A\mathbf{v}, \qquad A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix}.$$

3.3 Example — shear

Map. Horizontal shear: tilt the unit square into a parallelogram, $(x_1, x_2) \mapsto (x_1 + x_2, x_2)$.



3.3 Example — shear (matrix)

Find the matrix. Apply T to $\mathbf{e}_1$ and $\mathbf{e}_2$, then stack the outputs as columns:

$$T(\mathbf{e}_1) = \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad T(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \Longrightarrow \quad A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Check: $A\mathbf{e}_1 = \mathbf{e}_1$ (bottom edge fixed); $A\mathbf{e}_2$ tilts the vertical side.

3.3 Example — shear (decompose, transform, combine)

Take $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

1. Decompose: $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2$.

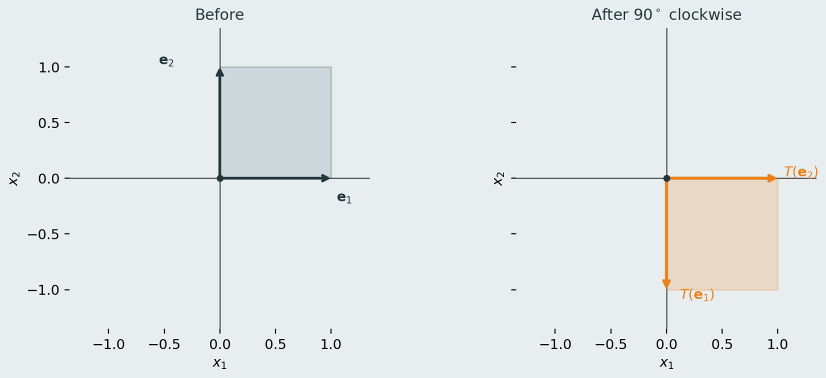
2. Transform: $2T(\mathbf{e}_1) = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad 3T(\mathbf{e}_2) = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$.

3. Combine:

$$T(\mathbf{v}) = 2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} = A\mathbf{v}.$$

3.3 Example — 90° rotation (clockwise)

Map. Rotate every vector 90° clockwise: $(x_1, x_2) \mapsto (x_2, -x_1)$.



3.3 Example — rotation (matrix)

Find the matrix. Apply T to $\mathbf{e}_1$ and $\mathbf{e}_2$, then stack the outputs as columns:

$$T(\mathbf{e}_1) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \qquad T(\mathbf{e}_2) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Check: $A\mathbf{e}_1$ points down; $A\mathbf{e}_2$ points right (see previous graph).

3.3 Example — rotation (decompose, transform, combine)

Same recipe with $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

1. Decompose: $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2$.

2. Transform: $2T(\mathbf{e}_1) = 2\begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad 3T(\mathbf{e}_2) = 3\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$.

3. Combine:

$$T(\mathbf{v}) = 2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = \begin{pmatrix} 0 \\ -2 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} = A\mathbf{v}.$$

3.3 Every linear map is a matrix

Claim. If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then there is a unique matrix $A \in \mathbb{R}^{m \times n}$ such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$.

Construction. Let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis of $\mathbb{R}^n$. Form A by stacking the outputs as columns:

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}.$$

Why it works. Any $\mathbf{x} = \sum_{j=1}^{n} x_j \mathbf{e}_j$. By linearity,

$$T(\mathbf{x}) = \sum_{j=1}^{n} x_j T(\mathbf{e}_j) = \sum_{j=1}^{n} x_j \mathbf{a}_j = A\mathbf{x}.$$

Column j of A is exactly where $\mathbf{e}_j$ is sent.

3.3 Null space — directions squeezed to 0

Setup ($A \in \mathbb{R}^{2\times 2}$). View A as a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(\mathbf{x}) = A\mathbf{x}$.

Null space / kernel (subspace of $\mathbb{R}^2$):

$$\mathrm{Null}(A) = \ker(T) = \{\mathbf{x} \in \mathbb{R}^2 : \ A\mathbf{x} = \mathbf{0}\}.$$

Vectors in $\mathrm{Null}(A)$ are mapped to $\mathbf{0}$ in the output.

Intuition. If $\mathbf{x} \neq \mathbf{0}$ lies in $\mathrm{Null}(A)$, the whole line $\mathrm{span}\{\mathbf{x}\} \subset \mathbb{R}^2$ is sent to the single point $\mathbf{0} \in \mathbb{R}^2$ — that **input direction is squeezed** (lost in the output).

Example ($\det A = 0$). $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: $A(1,-1)^{\mathsf{T}} = \mathbf{0}$, so $\mathrm{Null}(A) = \mathrm{span}\{(1,-1)^{\mathsf{T}}\}$ — one direction squeezed.

3.3 Rank–nullity — squeezed vs. persist (2×2)

Setup. $A \in \mathbb{R}^{2\times 2}$, $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(\mathbf{x}) = A\mathbf{x}$. The plane has 2 input dimensions; under T, each splits into one of two fates:

1. Squeezed to 0 (kernel, in $\mathbb{R}^2$): $\ker(T) = \mathrm{Null}(A)$; dimension $= \mathrm{nullity}(A)$.

2. Persist in the output (image, in $\mathbb{R}^2$): $\mathrm{Im}(T) = \mathrm{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^2\}$; dimension $= \mathrm{rank}(A)$ — independent directions that **survive** in the output (not collapsed to $\mathbf{0}$).

Rank–nullity. $\mathrm{rank}(A) + \mathrm{nullity}(A) = 2$.

Example ($\det A = 0$). $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: nullity $= 1$ (direction $(1, -1)^{\mathsf{T}}$ squeezed), rank $= 1$ (outputs fill the line $\mathrm{span}\{(1,1)^{\mathsf{T}}\}$).

Example ($\det A \neq 0$). Shear or rotation from earlier slides: nullity $= 0$ (only $\mathbf{0}$ squeezed), rank $= 2$ (both input dimensions persist — output still fills $\mathbb{R}^2$).

3.3 Rank–nullity — general case (for interest only)

Setup. Let $A \in \mathbb{R}^{m \times n}$, $T : \mathbb{R}^n \to \mathbb{R}^m$, $T(\mathbf{x}) = A\mathbf{x}$.

Kernel and image. $\ker(T) = \mathrm{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$;
$\mathrm{Im}(T) = \mathrm{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$.

Rank and nullity. $\mathrm{rank}(A) = \dim(\mathrm{Im}\ T)$, $\quad \mathrm{nullity}(A) = \dim(\ker T)$.

Rank–nullity theorem.

$$\mathrm{rank}(A) + \mathrm{nullity}(A) = n \qquad (n \text{ columns} = \dim(\mathbb{R}^n)).$$

Same intuition. Of the n domain dimensions, nullity(A) are **squeezed** to $\mathbf{0} \in \mathbb{R}^m$; rank$(A)$ independent directions **persist** in $\mathrm{Im}(T) \subseteq \mathbb{R}^m$. When $n = m$, invertible $\Leftrightarrow$ nullity $= 0 \Leftrightarrow$ rank $= n$.

3.3 After a squeeze — what survives? (2×2)

Recall. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ maps the whole plane onto one output line (through $(1,1)^\top$). Direction $(1,-1)^\top$ is squeezed to $\mathbf{0}$.

Question. Given an input $\mathbf{v}$, what is $A\mathbf{v}$?

Key idea (in words). Think of $\mathbf{v}$ as two moves: one along the **squeezed** direction (the map **forgets** it) and one along the surviving direction $(1,1)^\top$ (the map **remembers** only this). So $A\mathbf{v}$ depends on how much of $\mathbf{v}$ points along the surviving direction — not on the part that collapses.

Example. $(3,1)^\top = 2(1,1)^\top + 1(1,-1)^\top$: the $(1,-1)^\top$ part is lost, the $2(1,1)^\top$ part survives, so $A\mathbf{v} = 2(2,2)^\top = (4,4)^\top$.

3.3 Dot product — algebraic and geometric

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

Algebraic definition.

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i .$$

Geometric definition. Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta .$$

Equivalent. These two definitions always give the **same number**.

Video (intuition and proof). https://www.youtube.com/watch?v=LyGKycYT2v0

3.3 Linear regression — $\mathbf{y}$ and X

Data (n workers). Worker i has income y_i, age age_i, education educ_i.

Response vector.

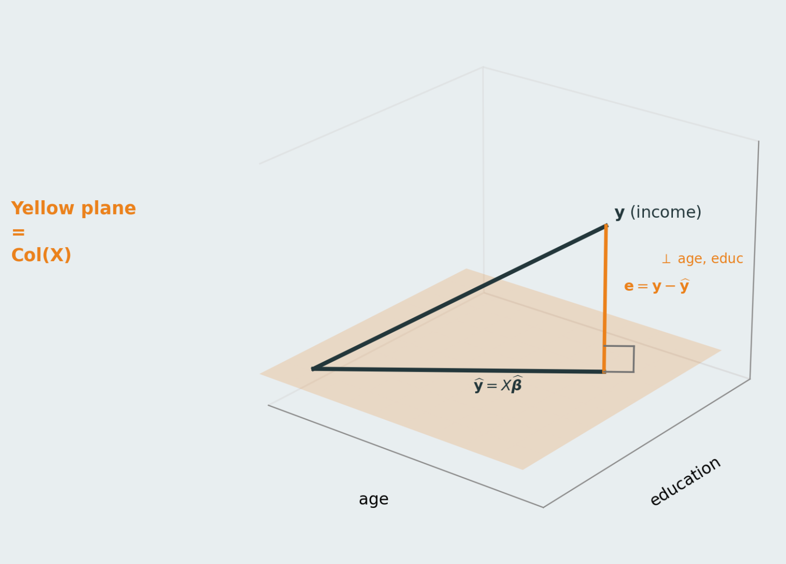
$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n.$$

Regressor matrix (two columns: age, educ).

$$X = \begin{pmatrix} \text{age}_1 & \text{educ}_1 \\ \vdots & \vdots \\ \text{age}_n & \text{educ}_n \end{pmatrix} \in \mathbb{R}^{n\times 2}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_{\text{age}} \\ \beta_{\text{educ}} \end{pmatrix}, \qquad \mathbf{y} \approx X\boldsymbol{\beta}.$$

Column j of X is the vector of worker i values for regressor j.

Schematic next ($n = 3$): age and education are the two axes; yellow plane $= \text{Col}(X)$; $\mathbf{e}$ is orthogonal to every column of X.


Yellow plane
=
Col(X)
y (income)
⊥ age, educ
e = y − ŷ
ŷ = Xβ̂
age
education

3.3 OLS fit — e orthogonal to every column of X

From the picture ($n = 3$). Yellow plane = Col(X); fitted $\widehat{\mathbf{y}} = X\widehat{\boldsymbol{\beta}}$ lies in it. Residual $\mathbf{e} = \mathbf{y} - X\widehat{\boldsymbol{\beta}}$ is perpendicular to the plane.

Orthogonal to every column of X. Let $\mathbf{x}_1$ = age column, $\mathbf{x}_2$ = educ column. Then

$$\mathbf{x}_1 \cdot \mathbf{e} = 0, \qquad \mathbf{x}_2 \cdot \mathbf{e} = 0 \qquad \Longleftrightarrow \qquad X^\top \mathbf{e} = \mathbf{0} \qquad \Longleftrightarrow \qquad X^\top\big(\mathbf{y} - X\widehat{\boldsymbol{\beta}}\big) = \mathbf{0}.$$

These **normal equations** characterise $\widehat{\boldsymbol{\beta}}$.

Solve for $\widehat{\boldsymbol{\beta}}$. $X^\top X\widehat{\boldsymbol{\beta}} = X^\top \mathbf{y}$. If the columns of X are independent (full column rank), then

$$\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

(Since $\widehat{\mathbf{y}} \in \mathrm{Col}(X)$, also $\widehat{\mathbf{y}} \cdot \mathbf{e} = 0$.)

3.3 OLS fit — same as least squares

Pythagoras in $\mathbb{R}^n$. Fix $\widehat{\boldsymbol{\beta}}$ with $\mathbf{e} = \mathbf{y} - X\widehat{\boldsymbol{\beta}}$. For any $\boldsymbol{\beta}$,

$$\mathbf{y} - X\boldsymbol{\beta} = \underbrace{\mathbf{y} - X\widehat{\boldsymbol{\beta}}}_{\mathbf{e}} + \underbrace{X(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})}_{\in \mathrm{Col}(X)}.$$

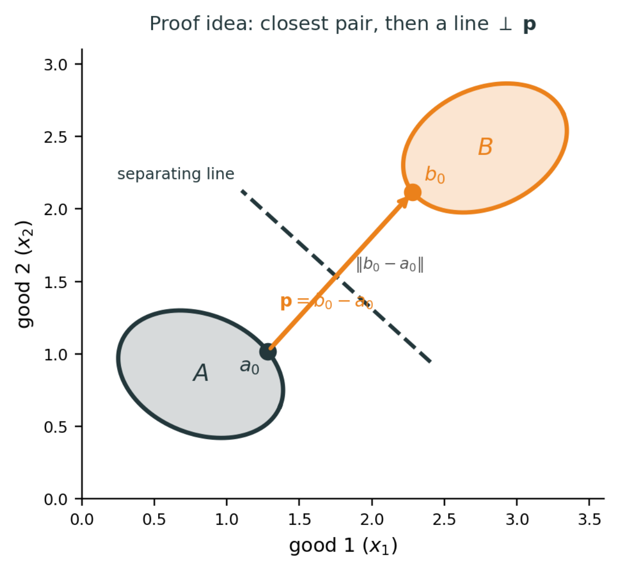
The two pieces are orthogonal, so

$$\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \|\mathbf{y} - X\widehat{\boldsymbol{\beta}}\|^2 + \|X(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \geq \|\mathbf{y} - X\widehat{\boldsymbol{\beta}}\|^2.$$

Conclusion. The smallest squared error occurs when $X(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{0}$, i.e. $\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(y_i - \beta_{\text{age}}\, \text{age}_i - \beta_{\text{educ}}\, \text{educ}_i\big)^2$.

Appendix

Not required for Math Camp. Proofs and extra examples linked from the main slides.



Proof idea: closest pair, then a line ⊥ p
separating line
B
b_0
$\|b_0 - a_0\|$
$\mathbf{p} = b_0 - a_0$
A
a_0
good 2 (x_2)
good 1 (x_1)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5

A.1 Proof. Separation when A is compact

for interest only

Claim. Disjoint nonempty convex $A, B \subset \mathbb{R}^n$ with A compact and B closed. Then $\exists\, \mathbf{p} \neq \mathbf{0}$, $\beta \in \mathbb{R}$ with $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ for all $a \in A$, $b \in B$.

Step 1 — closest pair. Define $d = \inf\{\|a - b\| : a \in A,\ b \in B\} > 0$. Pick sequences $(a_n) \subset A$, $(b_n) \subset B$ with $\|a_n - b_n\| \to d$. Since A is compact, a subsequence $a_{n_k} \to a_0 \in A$. The (b_{n_k}) are bounded, so (using B closed) a further subsequence $b_{n_{k_j}} \to b_0 \in B$. Hence the infimum is **attained** at some pair $(a_0, b_0) \in A \times B$.

Not an interior point. a_0 and b_0 need only lie in A and B; they are usually on the **boundaries**, not in $\mathrm{int}(A)$ or $\mathrm{int}(B)$.

Step 2 — normal vector. Set $\mathbf{p} = b_0 - a_0 \neq \mathbf{0}$ (points from a_0 toward b_0).

A.1 Step 1 — why the subsequence argument works

for interest only

Why $a_{n_k} \to a_0 \in A$. In $\mathbb{R}^n$, **compact** = closed + bounded (Heine–Borel). Every sequence in a compact set has a **subsequence** converging to a limit **in that set**. So $(a_n) \subset A$ compact $\Rightarrow\ \exists\, a_{n_k} \to a_0 \in A$.

Why (b_{n_k}) is bounded. Since $\|a_{n_k} - b_{n_k}\| \to d$ and (a_{n_k}) converges, for large k,

$$\|b_{n_k}\| \le \|b_{n_k} - a_{n_k}\| + \|a_{n_k}\| \le d + 1 + \|a_0\| + 1.$$

(Triangle inequality: $\|b\| \le \|b - a\| + \|a\|$.)

Why $b_{n_{k_j}} \to b_0 \in B$. A bounded sequence in $\mathbb{R}^n$ has a convergent subsequence (Bolzano–Weierstrass). Since B is **closed**, the limit b_0 lies in B.

A.1 Step 3 — why the derivative works

for interest only

Setup. Fix $a \in A$. For $t \in [0,1]$, convexity gives $a_t = a_0 + t(a - a_0) \in A$ and $g(t) = \|a_t - b_0\|^2$. Since (a_0, b_0) minimizes distance, $t = 0$ minimizes g on $[0,1]$.

Why use the derivative? A minimum at the left endpoint $t = 0$ means the slope cannot point downward: $g'(0) \geq 0$.

Differentiate. $g(t) = \|a_0 + t(a - a_0) - b_0\|^2$, so

$$g'(t) = 2\langle a_0 + t(a - a_0) - b_0,\ a - a_0\rangle, \qquad g'(0) = 2\langle a_0 - b_0,\ a - a_0\rangle \geq 0.$$

Hence $\langle b_0 - a_0,\ a - a_0\rangle \leq 0$. With $\mathbf{p} = b_0 - a_0$: $\mathbf{p} \cdot (a - a_0) \leq 0$, so $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$.

A.1 Proof. Separation when A is compact (continued)

Step 4 — B stays on the other side. Fix $b \in B$. For $t \in [0,1]$, set $b_t = b_0 + t(b - b_0) \in B$ and $h(t) = \|a_0 - b_t\|^2$. Since $t = 0$ minimizes h on $[0,1]$, the same derivative argument gives $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$ for all $b \in B$.

Step 5 — finish. Since $A \cap B = \varnothing$, the closest points satisfy $a_0 \neq b_0$. With $\mathbf{p} = b_0 - a_0$,

$$\mathbf{p} \cdot b_0 - \mathbf{p} \cdot a_0 = \mathbf{p} \cdot (b_0 - a_0) = \|b_0 - a_0\|^2 = d^2 > 0,$$

so $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$. Any β with $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$ works (e.g. $\beta = \mathbf{p} \cdot a_0$ gives **weak** separation: $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ for all $a \in A$, $b \in B$).

A.1 Strict separation

for interest only

Weak vs. strict. From Steps 3–5: $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$ and $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$. So any β with $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$ separates weakly (e.g. $\beta = \mathbf{p} \cdot a_0$).

When is separation strict? If $d = \|b_0 - a_0\| > 0$ (sets bounded apart; e.g. both compact and disjoint), then $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$ with a **positive gap**. Pick β **strictly between** them, e.g. $\beta = \frac{1}{2}(\mathbf{p} \cdot a_0 + \mathbf{p} \cdot b_0)$. Then

$$\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0 < \beta < \mathbf{p} \cdot b_0 \leq \mathbf{p} \cdot b \quad \Rightarrow \quad \mathbf{p} \cdot a < \beta < \mathbf{p} \cdot b.$$

If $d = 0$ (sets touch in the limit, e.g. $A = [0, 1]$, $B = (1, 2]$), only weak separation is possible.

A.1 Proof. Remarks

for interest only

If only B is compact. Swap the roles of A and B, apply the proof, then relabel the normal vector.

Single point $B = \{b_0\}$. Let $\hat{a}$ be a point in A closest to b_0 (exists since A is closed and convex). The same calculation with $(a_0, b_0) = (\hat{a}, b_0)$ separates A from b_0 even when A is not compact.

A.2 Proof. Efficient production $\Rightarrow$ profit maximization

for interest only

Goal. $y \in Y$ technically efficient $\Rightarrow\ \exists\, \mathbf{p} \neq \mathbf{0}$ with $\mathbf{p} \cdot y \geq \mathbf{p} \cdot y'$ for all $y' \in Y$.

Assumptions (camp level). $Y \subset \mathbb{R}^n$ is **closed and convex**; Y is **bounded** (so compact); **free disposal** (if $y \in Y$ and $\tilde{y} \leq y$, then $\tilde{y} \in Y$).

Step 1 — better set. Define

$$P_y = \{y' \in \mathbb{R}^n : \ y' \geq y,\ y' \neq y\}$$

(weakly more output, strictly more in at least one good). P_y is **convex**.

Step 2 — disjointness. If $y' \in P_y \cap Y$, then y' weakly dominates y in Y, contradicting efficiency. So $P_y \cap Y = \varnothing$.

A.2 Proof. Efficient production (continued)

for interest only

Step 3 — separate. Y is compact and P_y is closed, so apply the separating hyperplane theorem (swap labels if needed so a compact set is on one side). There exist $\mathbf{p} \neq \mathbf{0}$ and β with

$$\mathbf{p} \cdot y'' \leq \beta \leq \mathbf{p} \cdot y' \quad \forall\, y'' \in Y,\ \forall\, y' \in P_y.$$

Hence $\mathbf{p} \cdot y'' \leq \mathbf{p} \cdot y'$ for all $y'' \in Y$, $y' \in P_y$.

Step 4 — profit maximization at y**.** For each good i, pick $y'^{(i)} = y + \varepsilon e_i \in P_y$ for small $\varepsilon > 0$ (feasible with free disposal). Then $\mathbf{p} \cdot y'' \leq \mathbf{p} \cdot y'^{(i)} = \mathbf{p} \cdot y + \varepsilon p_i$. Let $\varepsilon \downarrow 0$ to get $\mathbf{p} \cdot y'' \leq \mathbf{p} \cdot y$ for all $y'' \in Y$.

Remark. Separation gives only $\mathbf{p} \neq \mathbf{0}$. For $\mathbf{p} \geq 0$, need free disposal on all goods: if $p_i < 0$, then $y - \varepsilon e_i \in Y$ (since $\leq y$) raises profit, contradicting maximization at y.